



Dilip Sahu
Roll No.: BT24CS028
B.Tech
Computer Science and Engineering
National Institute Of Technology,
Mizoram



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CAREER OBJECTIVES

Aspiring Software Development Engineer with expertise in designing scalable, fault-tolerant architectures, low-latency systems, and cloud-native applications. Proficient in Agile methodologies and solving complex technical challenges to deliver innovative, high-impact solutions.

EDUCATION

Degree	Institute	Board / University	CGPA/Percentage	Year of passing
B.Tech	National Institute of Technology, Mizoram	NIT Mizoram	7.98 (Till 4th Sem)	2028
XII	Impulse Academy, Jaipur	RBSE	86.00%	2023

INTERNSHIPS

- **Software Engineer Intern** **Duration:** [Jul 2026 - Aug 2026](Remote)
Zetheta Algorithms Pvt. Ltd.
 - Engineered **FastAPI** reconciliation engine processing 100,000+ multi-bank transactions using **RapidFuzz**.
 - Built automated data ingestion pipelines supporting 3+ complex banking formats via **Pydantic**.
 - Deployed tamper-evident **PostgreSQL** audit databases with **SHA-256**, ensuring 100% transaction traceability.

PROJECTS

1. **CyberGuard-AI - A Real Time Scam Detector** **Github**
Android Application & Edge AI Deployment
 - Implemented **O(1) Bloom Filter** for instant, memory-efficient known-caller and spam **database verification**.
 - Engineered a **finite state-machine graph tracking** conversational phases to detect scam narrative arcs.
 - Designed a temporal **sliding window** algorithm to maintain continuous linguistic context for NLP.
2. **Coding & AI Club NIT Mizoram Website** **Website** **GitHub Link**
Backend System Design & Full Stack Development
 - Optimized registration pipelines utilizing **Hash Sets**, reducing duplicate validation time complexity to **O(N)**.
 - Engineered **exponential backoff algorithms** with jitter for highly resilient, fault-tolerant **API data synchronization**.
 - Implemented **DOM tree traversal** algorithms for hover-intent link prefetching, achieving near-zero ms latency.
 - Decreased form submission **latency by 95% (from 2000ms to <100ms)** utilizing asynchronous multithreading.
3. **IoT Based Bus-Tracking System** **Website** **GitHub Link**
Backend System Design & Engineering
 - Scaled ingestion to **10,000+ req/sec** using **Node.js**, **Redis FIFO queues**, and **SSE**.
 - Handled **500+ baseline pings** and **5,000+ peak req/sec** using **Haversine ETA algorithm**.
 - Reduced map update **latency to <50ms** using **Leaflet graph bounds** and **PostgreSQL**.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, HTML, CSS, JavaScript
- **Tools and Frameworks:** Visual Studio Code, PyCharm, GitHub, Bootstrap
- **MERN Stack:** MongoDB, Express.js, React, Node.js
- **Databases and Storage Systems::** MongoDB, MySQL
- **Operating Systems:** Windows, Linux
- **Frameworks Methodologies:** Agile Methodology, Cloud-Native Applications
- **System Design:** Scalable Architecture, Low-Latency Systems

KEY COURSES TAKEN

C++ Programming, Data Structure and Algorithm, Object-Oriented Programming (OOP), Computer Organization and Architecture, DBMS, Cloud Computing, Distributed Systems, Operating System, Computer Networks, Probability and Stochastics, Compiler Design, Computer Graphics, Artificial Intelligence, Machine Learning

CERTIFICATIONS

- **C++ Programming** Geeks for Geeks
- **React Web Developer Certification** HackerRank
- **Software Engineering** JP Morgan Chase Co. & Deloitte
- **Drone Workshop BootCamp** JetAerospace & IIT Delhi
- **Machine Learning for Data Science Projects** IBM

POSITIONS AND RESPONSIBILITIES

- **Vice President** Coding & AI Club, NIT Mizoram July 2026 - Present
- **Website & Media Coordinator** Training and Placement Cell, NIT Mizoram May 2026 - Present
- **Ex- Team Head - Website & Social Media Team** Coding & AI Club NIT Mizoram Feb 2025 - July 2026

ACHIEVEMENTS

- **National Finalist - 5G Innovation Hackathon:** Selected among the top 50 teams nationwide by the Department of Telecommunications for engineering **CyberGuard-AI**, an edge-intelligence security solution. 2026
- **Prototype Development Funding - Ministry of Communications:** Awarded a seed grant of **1,00,000** to accelerate prototype development and hardware integration for CyberGuard-AI ahead of the India Mobile Congress. 2026
- **1st Prize - ARTHKRITI AI Hackathon:** Engineered **TourAI**, a multi-module predictive travel planner application, securing first position in the technical build competition. 2026
- **1st Prize in IDEA Pitching - IIC & ArthKriti:** Conceptualized a UI-friendly platform for Mizoram Police featuring automated FIR routing, complaint tracking, and digital proof uploads. 2026
- **Solved 550+ Questions on LeetCode & Striver's Sheet:** Mastered a diverse range of algorithmic patterns across easy, medium, and hard difficulty levels to optimize problem-solving proficiency.
- **Shortlisted for Advanced Evaluation:** Amazon Machine Learning Summer School, Polaris Fellowship, HPCL-Mittal Energy Quest 3.0. 2026

HOBBIES

- **Competitive Programming**